Comparative Analysis of Short-Term Load Forecasting Methods

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***Abstract*— One of the primary tasks of electric utilities is to accurately predict the load demand requirements of consumers, especially for short term prediction. In view of this, different methods have been proposed for load prediction. In this paper, three methods (i.e. Multiple Linear Regression (MLR), Seasonal Auto Regressive Integrated Moving Average with Exogenous Variables (SARIMAX) and Long Short Term Memory (LSTM)) are compared to forecast load consumption in a typical Nigerian University. The main objective is to determine which of the techniques best model the load consumption pattern of the University accurately. Load forecast was made for weekdays (Monday-Friday) and weekends (Saturday and Sunday). The result showed that the LSTM technique is the best performing model achieving the least errors. The technique returns the mean absolute error (MAE) that varies between 0.029-0.093, mean square error (MSE) ranging between 0.0014-0.014 and root mean square error that has values between 0.037-0.12.**

***Index Terms*—Comparative analysis, Short Term Load Forecast, Load consumption, University of Ibadan**

I. INTRODUCTION

Forecasting is the process of making statements about events whose actual outcomes have not yet been observed. It is fundamental for decision-making in energy distribution [1]. A power system must be well planned to ensure an adequate and reliable power supply to meet the load demand. An important precondition for system planning is to arrive at a realistic estimation for future load demands [2]. Circumstances such as population burst, extreme weather, equipment failure or vandalism, etc., can cause energy demand on power stations to become strained leading to epileptic, inadequate, and unreliable power supply. Application of Load Forecasting in such situations can help in the optimization of important decisions such as load switching, voltage control, network reconfiguration, infrastructure development, scheduling of start-up and shut-down times of generating units, planning an overhaul, load management, etc. [1]. The period of forecast is used to classify load forecasting; short term, medium term and long term. Short-term forecast ranges from one hour to one week and aims at predicting electric loads for a period of minutes, hours, days, or weeks [1].

A universally best technique for load forecasting, whether short, medium, or long term, does not exist as there is no single forecast that can satisfy all needs. This is due to factors that affect energy consumption such as climate, and human activities, etc., which are not the same everywhere [3]. The common practice is to use different forecasts for different purposes or geographical locations, allowing the characteristics of the load data and the specific use to determine the choice of the most suitable and appropriate technique to be used [3]. Several methods and techniques can and have been used to perform load forecasting [4]. This work aims to use three popular models for short term load forecasting to forecast load demand one week in advance and then make a comparison. The Models are: (1) Multiple Linear Regression (MLR), (2) a Time Series Model, Seasonal Auto Regressive Integrated Moving Average with Exogenous Variables (SARIMAX) and (3) Long Short Term Memory (LSTM). This is achieved by presenting a study case using the distribution data of the power station of the University of Ibadan (UI), in Southwestern Nigeria. The electricity supply to the campus is powered via two dedicated priority lines through the National grid. Demographics, econometrics, and weather conditions are usually needed to be considered during load forecasting. However, the models used for this research will be trained without such factors as inputs. This is due to the inaccuracies of weather forecasts and the non-readily availability of information on such factors at the point of need. Also, the University of Ibadan has varying temperatures with the different regions of Nigeria as well as the fact that temperature changes constantly.

II. MULTIIPLE LINEAR REGRESSION, TIME SERIES AND LONG SHORT TERM MEMORY

A. Multiple Linear Regression (MLR)

Multiple linear regression models the relationship between several independent variables and a dependent variable as a

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linear function. A multiple linear regression with k number of independent variables, can be written as

$$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_k X_k + \varepsilon, \quad (1)$$

where y is the dependent variable (electric load), $X_1 \ldots X_k$ are independent variables (e.g., temperature, humidity and time of the day), β is regression coefficient, and ε is the error term.

B. Time Series: (Seasonal) Auto Regressive Integrated Moving Average (with Exogenous Variables) (SARIMAX)

Time Series methods is a popular approach used for short-term load forecasting. A model is initially created based on past data then the future load is predicted with the developed model [2]. Time Series Methods include [4]: (1) Autoregressive Moving Average (ARMA), (2) Autoregressive Integrated Moving Average (ARIMA), (3) ARIMA with Exogenous Variables (ARIMAX), (4) Seasonal ARIMA (SARIMA), and (5) SARIMA with Exogenous Variables (SARIMAX).

ARMA and ARIMA are combinations of Autoregressive (AR) and Moving Average (MA) models and use time and load as the only input parameters. An AR is a time series linear model and AR(1) means an AR of order 1. The order indicates the number of time lags used for the present time forecast. AR(1) is given by

$$y_t = \delta + \varphi y_{t-1} + w_t, \quad (2)$$

and AR(2) is given by

$$y_t = \delta + \varphi_1 y_{t-1} + \varphi_2 y_{t-1} + w_t. \quad (3)$$

where δ is a constant, φ is AR coefficient and w_t is the error (or white noise). MA (Moving Average) makes use of past forecast errors in a regression-like model. MA(1) is given by

$$y_t = \mu + w_t + \theta_1 W_{t-1}, \quad (4)$$

and MA(2) is given by

$$y_t = \mu + w_t + \theta_1 w_{t-1} + \theta_2 w_{t-2}, \quad (5)$$

where μ is a constant, θ is MA coefficient and w_t is the error.

ARMA is used for stationary time series data. Stationary data means that the data's properties like mean, variance, covariance etc. do not change with time [4]. With ARMA, the present value of y_t is dependent on its past values (i.e. y_{t-1}, y_{t-2}, etc.), and also ion that of white noise (w_{t-1}, w_{t-2}, etc) [10]. An ARMA of order (p, q), (where p and q are the orders of the AR and MA components respectively) is given by [10]

$$y_t = \varphi_1 y_{t-1} + \ldots + \varphi_p y_{t-p} + w_t - \theta_1 w_{t-1} - \ldots - \theta_q w_{t-q}. \quad (6)$$

ARIMA is used for non-stationary time series data. If non-stationary, the data must be transformed into stationary data by differencing [10]. Differencing introduces the ∇ operator, forming [10]

$$\nabla y(t) = (1 - B)\, y(t). \quad (7)$$

For data that needs to differenced d number of times with AR and MA orders p and q, the ARIMA (p, d, q) model is written as

$$\phi(B)\, \nabla^d y_t = \theta(B)\, w_t, \quad (6)$$

while a SARIMA model with order (p, d, q)x(P, D, Q), which is created by implementing an ARIMA model on data with seasonal variations, is represented as

$$\phi(B)\, \Phi(B^s)\, \nabla d\, \nabla_s^D yt = \theta(B)\, \varnothing(B^S) w_t, \quad (7)$$

where P is the order of the seasonal AR, D is the order of the seasonal differencing and Q is the order of seasonal MA. B is a backshift operator and is given by

$$B y_t = y_{t-1}. \quad (8)$$

$\phi(B)$ is the nonseasonal AR backshift operator and is given as

$$\phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \ldots - \phi_p B^p. \quad (9)$$

$\theta(B)$ is the nonseasonal MA backshift operator and is given as

$$\theta(B) = 1 - \theta_1 B - \theta_2 B^2 - \ldots - \theta_q B^q. \quad (10)$$

$\Phi(B^s)$ is the seasonal AR backshift operator and is given as

$$\Phi(B^s) = 1 - \Phi_1 B^s - \Phi_2 B^{2s} - \ldots - \Phi_P B^{Ps}. \quad (11)$$

$\varnothing(B^S)$ is the seasonal MA backshift operator and is given as

$$\varnothing(B^S) = 1 - \varnothing_1 B^s - \varnothing_2 B^{2s} - \ldots - \varnothing_Q B^{Qs}. \quad (12)$$

∇^d (= $[1 - B]^d$) and ∇_s^D (= $[1 - B^s]^D$) are the non-seasonal and seasonal difference operators of order d and D respectively.

For obtaining a more accurate forecast, ARIMA can be incorporated with exogenous variables that correlate with the historical data. ARIMA (p, d, q) becomes ARIMAX (p, d, q) and is given by

$$\phi_p(B) y_t = \theta_q(B) w_t + v_i(B) X_{t-i} \quad (13)$$

Where v_i is the coefficient of the i-lagged exogenous input, X. Seasonal ARIMA (SARIMA) can also be further coupled with exogenous variables to give a SARIMAX model.

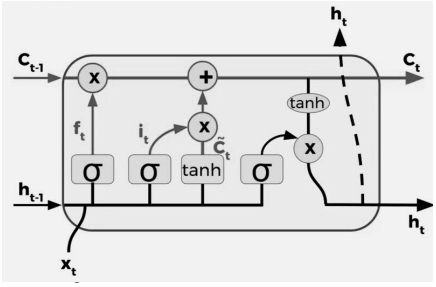
C. LONG SHORT TERM MEMORY (LSTM)

Machine learning algorithms such as Artificial Neural Networks and Support Vector Machines is another very popular methodology used for load forecasting. Flexible and capable of handling complexity and non-linearity, machine learning algorithms have an advantage over more traditional load forecasting methods like Time Series [13].

Artificial Neural Networks (ANN) are designed to imitate the way the human brain handles information. ANN composes of nodes, called neurons, with which it receives information (through input nodes) and sums up the information. With the help of an activation function, the neuron then transmits its response to other neurons in the Network. The performing capability of the neural network is directly dependant on the strength of the connections between the nodes, called weights [4]. Weights are obtained by learning or adapting from a set of training patterns [4]. ANN can be assigned into different categories based on supervised and unsupervised learning methods and feed-forward and feedback recall architectures [4].

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LSTM is a special kind of ANN called Recurrent Neural Networks (RNN). LSTM and RNN are sequence-based models which can maintain temporary correlations between past data points and current data point. This means that the decision(s) taken at time step t, will be affected by the decision(s) made at time step t-1, and likewise, the decision(s) to be made at t+1 will be affected by the decision(s) made at time t. This enables them to learn trends that are associated with human behaviour better than traditional feed-forward neural networks [13]. However, RNNs experiences difficulty in learning long-range dependencies, an issue LSTM was developed to solve [14].



From Fig. 1, C_t is the memory cell state, h_t is the hidden state (which is the memory of the network) and x_t is the current time step input. i, f and o are called the input, forget and output gates, respectively (they are called gates because they interact with sigmoid functions which limits the values of the vectors between 0 and 1). LSTM function as such: (1) the forget gate layer, where the model decides which information to throw away or forget from the cell state,

$$f_t = \sigma(W_f * [h_{t-1}, x_t] + b_f), \quad (14)$$

(2) the model then decides which new information to store in the cell state,

$$i_t = \sigma(W_i * [h_{t-1}, x_t] + b_f), \quad (15)$$

$$C'_t = tanh(W_c * [h_{t-1}, x_t] + b_c) \quad (16)$$

(3) the model updates the old cell state,

$$C_t = f_t * C_{t-1} + i_t * C't \quad (17)$$

(4) and finally decides what to output for h_t.

$$o_t = \sigma(W_o * [h_{t-1}, x_t] + b_o), \quad (18)$$

$$h_t = o_t * tanh(C_t) \quad (19)$$

W_f, W_i, W_c and W_o are weight matrices for the corresponding inputs of the network activation functions, $*$ stands for an element-wise multiplication and σ represents the sigmoid activation function.

III. DATA ANALYSIS AND EXPERIMENT

The selected methods were implemented on hourly time series electric load data for a period of 6 months (June 2019 to November 2019) which was obtained from the power station of the University of Ibadan (UI), Ibadan. The data were transferred to Microsoft Excel and periods with power outages from the National grid were recorded as zero loads. Fig. 2 shows the total load demand during the sampling period which summed up to be approximately 5.2 x 10^3 MW.

As seen in, Fig. 3, the load demand rises during the weekends and starts to reduce at the start of the week on Monday before picking up just at the start of the weekends on Thursday. This can as a result of an increase in activities over the weekends, from religious activities to parties and ceremonies, as well as, more residents (students and staff) being home. Fig. 4 shows that the peak demand hours are between the times 9 a.m. and 2 p.m. This timeframe corresponds to working hours (students in their lecture halls and departments, lecturers and other staff in their offices and vendors and like offering services) which could serve as an explanation for the increased load demand. The load demand drops by 3 p.m. and rises again by 7 p.m. which corresponds to most individuals (students and staff) back in their homes (and hostels). There is a drop in demand again by 11 p.m. when retirement for the night commences and it rises again by 6 p.m. before starting to peak by 9 p.m.

As there is no general rule for the selection of the number of input variables [15], after several trials, previous hour load and previous week load were chosen as input variables as the models performed better with the selection. The dataset was divided into three sets namely: (1) Training Data: Data from 1st of June 2019 to 16th of November 2019, (2) Validation data: Data from 17th of November 2019 to 23rd of November 2019 (3) Test data: Data from 24th of November 2019 to 30th of November 2019.

A. Performance Measurement

There are many ways to assess the accuracy of a forecast. Three measures were used to evaluate the performance of the

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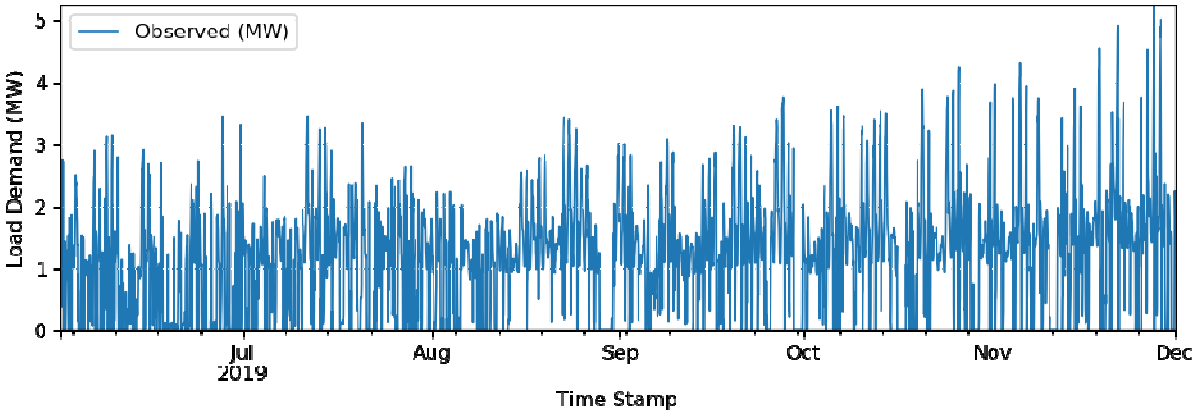


Figure 2. Load Demand Monthly Plot

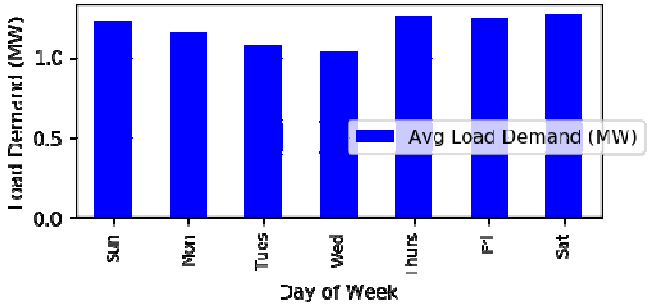


Figure 3. Average Weekly Load Consumption Rate

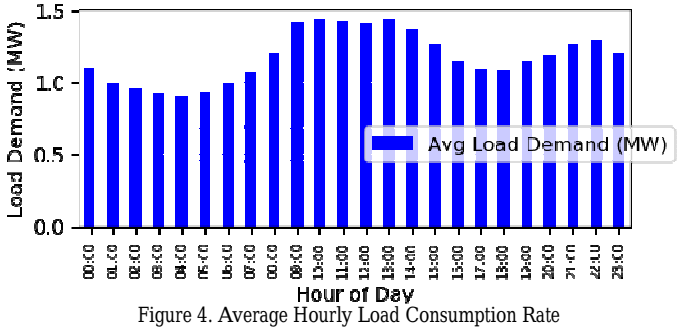


Figure 4. Average Hourly Load Consumption Rate

models presented in this work; (1) Mean Absolute Errors (MAE) (2) Mean Squared Error (MSE) (3) Root Mean Squared Error (RMSE). Generally, the smaller the error of any evaluation metric used, the better the models performs.

- Mean Absolute Errors:

$$\frac{1}{n}\sum_{i=1}^{n}|f_i - a_i| \tag{20}$$

where $f = (f_1, \ldots, f_n)$ is the forecast and $a = (a_1, \ldots, a_n)$ is the actual (observed) load.

- Mean Squared Error:

$$\frac{1}{n}\sum_{i=1}^{n}(f_i - a_i)^2 \tag{21}$$

- Root Mean Squared Error:

$$\sqrt{\frac{1}{n}\sum_{i=1}^{n}(f_i - a_i)^2} \tag{22}$$

The programming language Python was used to conduct the study. Python contains numerous packages, libraries and modules, some of which were used for the forecasts. The Scikit learn (sklearn) library was used for the implementation of the MLR model and forecasting. Sklearn was also used for the evaluation metrics across all the models.

For the SARIMAX model, Statmodels and some of its related modules were used for implementation. For the selection of parameters for the SARIMAX model, (p, d, q, P, D, Q), a correct interpretation of the autocorrelation function (ACF) plot, the partial autocorrelation function (PACF) plot is normally necessary. However, the Python package Auto-Arima, which takes into account the Bayesian information criterion (BIC) and the Akaike information criterion (AIC) to determine the best combinations of parameters automatically, is used to identify the most suitable model parameters. The parameters obtained were (3, 0, 1) for the non-seasonal orders (i.e. p, d, q) and the for seasonal orders, $(2, 0, 1)_{24}$, with 24 representing the seasonality

The LSTM Model was implemented with the Keras Library. After experimentation and several fine-tuning, the LSTM is trained with the following layer specifications: (1) Input features = 2, (2) Output /Dense layer = 1, (3) No. of Hidden LSTM Units = 350, (4) Solver = adam, (5) No. of Training Iterations (MaxEpochs) = 250, (6) Activation = 'relu'.

IV. RESULTS

Shown below is the obtained MLR, SARIMAX, and LSTM forecasts across the week from the 24th, November 2019 to 30th, November 2019. Fig. 5 shows the forecasts for the weekdays, Monday to Thursday, while Fig. 6 shows the forecasts for the weekends, Friday to Sunday.

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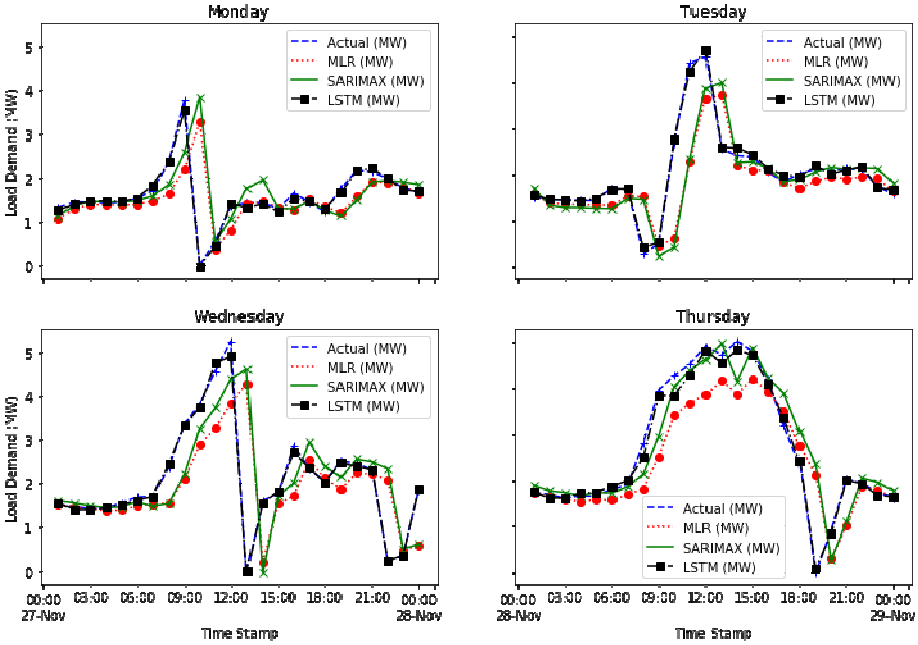


Figure 5. Forecasts for the Weekdays

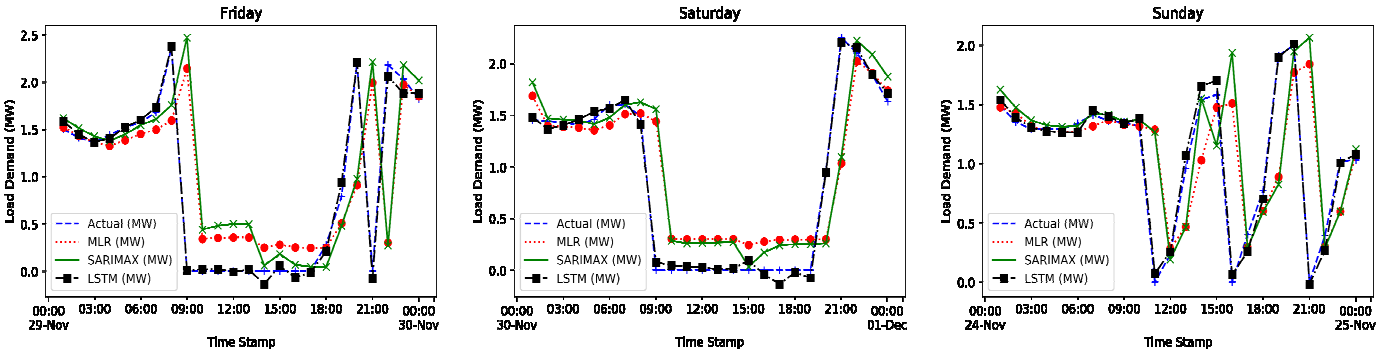


Figure 2. Forecasts for the Weekend days

Table 1. Comparison of Results

Day of the week	MLR			SARIMAX			LSTM		
	MAE	*MSE*	*RMSE*	*MAE*	*MSE*	*RMSE*	*MAE*	*MSE*	*RMSE*
24/11/2019 Sunday	0.34	0.38	0.62	0.42	0.51	0.72	0.06	0.01	0.07
25/11/2019 Monday	0.39	0.63	0.79	0.43	0.76	0.87	0.06	0.07	0.08
26/11/2019 Tuesday	0.43	0.55	0.74	0.41	0.58	0.70	0.07	0.01	0.09
27/11/2019 Wednesday	0.74	1.41	1.19	0.71	1.66	1.29	0.06	0.01,	0.10
28/11/2019 Thursday	0.55	0.58	0.76	0.47	0.49	0.70	0.11	0.02	0.15
29/11/2019 Friday	0.48	0.64	0.80	0.45	0.76	0.86	0.05	0.01	0.07
30/11/2019 Saturday	0.30	0.21	0.46	0.23	0.21	0.46	0.05	0.003	0.06

Table 1. compares the errors of the three models forecasts across the week. It can be seen that the model with the least error from Sunday 24th, November 2019 to Saturday 30th, November 2019 is the LSTM model indicating that it is the best fitting model. The LSTM not only outperformed but also gave a highly accurate forecast, unlike the MLR and SARIMAX which performed badly as seen from Fig. 5. It can also be observed from Fig. 5 that the SARIMAX model performed better at the beginning of each day, but as the day progressed, the energy consumption became more difficult for the SARIMAX model to forecast.

V. CONCLUSION

This work has investigated three different methods for short-term load forecasting applications in Nigeria. A real Nigeria utility's load distribution data was used. These results were obtained with the following input variables; the previous hour load and previous week load. Due to having the least errors, the LSTM model was observed to be the best performing model compared to MLR and SARIMAX models.

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Future work can incorporate information about temperature and weather as additional exogenous variable if possible as that should produce even better-performing models.

VI. ACKNOWLEDGMENT

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VII. REFERENCES

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